



3D Printing Modelling Engineer

QP Code: SSC/Q8903

Version: 3.0

NSQF Level: 5

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Qualification Pack

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SSC/Q8903: 3D Printing Modelling Engineer

Brief Job Description

Individuals at this job are responsible for creating design, conceptualization, development of 3d models in the design software programs and generation of the G-codes for printing the product

Personal Attributes

This job requires the individual to have thorough knowledge of various technology trends and processes as well as have updated knowledge about IT initiatives. He/she should be highly motivated and energetic with the ability to self-direct daily activities.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SSC/N8912: Design and develop finalized 3D models and product prototypes using CAD/CAM programs](#)
2. [SSC/N8913: Structure and conduct appropriate simulation tests, design optimization techniques to validate and finalize the 3D CAD models.](#)
3. [SSC/N8914: Export print-ready digital design files and generate tools paths for 3D printers in the form of G-code instructions](#)
4. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	3D Printing
Country	India
NSQF Level	5
Credits	16
Aligned to NCO/ISCO/ISIC Code	NCO-2015/2513.NIL

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Minimum Educational Qualification & Experience	Completed 2nd year of UG (UG Diploma) (UG/PG/degree/diploma with courses related to Engg./ Science) OR Completed 3 year diploma after 10th (UG/PG/degree/diploma with courses related to Engg./ Science) with 1.5 years of experience in relevant field. Relevant experience in job roles related to 3D Printing/ Additive Manufacturing The relevant experience would include work, internship and apprenticeship undertaken post completion of relevant educational qualification. OR Previous relevant Qualification of NSQF Level (4.5) with 1.5 years of experience in relevant field. Relevant experience in job roles related to 3D Printing/ Additive Manufacturing The relevant experience would include work, internship and apprenticeship undertaken post completion of relevant educational qualification.
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NIL
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	07/10/2028
NSQC Approval Date	07/10/2025
Version	3.0
Reference code on NQR	QG-05-IT-01509-2025-V2-NASSCOM
NQR Version	2

Remarks:

NA

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SSC/N8912: Design and develop finalized 3D models and product prototypes using CAD/CAM programs

Description

This unit is about incorporation of DfAM principles to optimize the design methodology and process optimization

Scope

The scope covers the following :

- Fundamentals of 3D Printing and CAD Design, Design Principles and Optimization Techniques, Mechanical Properties and Strength Analysis, Printing Strategy and Support Optimization, Software Tools and Automation, Creative Design Elements, Market and User Considerations, Quality and Compliance Standards

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Incorporate DfAM (Design for Additive Manufacturing) principles to optimize the design methodology based on 3D printing technology.
- PC2.** Evaluate part consolidation, geometric freedom, and biomimicry to enhance design capabilities.
- PC3.** Consider key process-specific, material-driven design guidelines, rules, and considerations.
- PC4.** Consider limitations of 3D printing parts, pre-defined design allowables in line with industry standards to comprehend design feasibility
- PC5.** Include suggested material types, colors, infill levels, orientations, and finishing methods in the 3D design.
- PC6.** Evaluate the constraints in physical properties of materials, including supports for overhangs, walls, and corners
- PC7.** Determine shapes of components with fewer stress points, smaller footprints, and superior mechanical properties
- PC8.** Decide on the minimum level of detail required based on the specific 3D printing technology.
- PC9.** Quantify and prioritize necessary mechanical properties, standards, and loads for the parts.
- PC10.** Refer to industry standards for applicable mechanical properties, including tensile, compression, flexural strength, and wear resistance
- PC11.** Analyze loading conditions to design parts such that largest forces traverse planes parallel to the print bed for optimal strength
- PC12.** Identify critical dimensions and features, including minimum wall thickness and support variables, adjusting for printer precision
- PC13.** Maximize print bed contact to minimize the need for supports and improve bed adhesion.
- PC14.** Orient the part optimally so the largest face lies on the print bed unless dictated otherwise by strength or geometry

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- PC15.** Reduce supports and improve overhangs by optimizing angles to decrease printing and processing time
- PC16.** Avoid large flat surfaces and use rounded corners to prevent warping during printing.
- PC17.** Use fillets or chamfer edges to ensure smooth transitions and reduce stress concentrationS
- PC18.** Thicken vulnerable areas and appendages from the core to prevent breakage
- PC19.** Conduct operations between software systems to enhance workflows and design capabilities
- PC20.** Compare CAD modeling software solutions to identify the most suitable package for 3D printing technology
- PC21.** Incorporate AI-driven generative design methodologies and software tools for innovative solutions
- PC22.** Set part parameters like mount points and environmental stresses without strictly dictating the shape of the object
- PC23.** Extract and modify geometries generated by AI-based algorithms in CAD systems
- PC24.** Design and create special effects, including particle effects and dynamic simulations
- PC25.** Utilize software solutions to automate design workflows through scripting
- PC26.** Define target user needs and product requirements to ensure alignment with market expectations
- PC27.** Analyze product lifecycle and sustainability, considering recyclability and environmental impact.
- PC28.** Use feedback from prototype testing and user trials to iteratively improve product design.
- PC29.** Establish quality assurance protocols for testing prototypes to ensure functionality and compliance.
- PC30.** Ensure alignment with branding and aesthetic considerations to create products that resonate with target audiences
- PC31.** Establish quality assurance protocols for testing prototypes to ensure functionality and compliance with specifications
- PC32.** Ensure alignment with industry standards for applicable mechanical properties.
- PC33.** Plan for manufacturability (DfM) by considering cost-effectiveness during production

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Organizational policies, procedures and guidelines which relate to evaluating and developing new use cases
- KU2.** Different data sources and how to access documents and information from data sources
- KU3.** Organizational policies and procedures for documenting new developed business cases
- KU4.** Who to involve while evaluating the feasibility of new business use cases and development plans
- KU5.** The range of standard templates and tools available and how to use them
- KU6.** Knowledge of different 3D software
- KU7.** Knowledge of Dfam principles
- KU8.** Understand AI applications in the 3D printing ecosystem

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KU9. Understand design automation workflows

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** analyze business impact and disseminate relevant information to others
- GS2.** apply problem-solving approaches in different situations
- GS3.** seek clarification and advice on problems from appropriate people
- GS4.** provide opinions on work in a detailed and constructive way
- GS5.** work effectively in a customer facing environment

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Incorporate DfAM (Design for Additive Manufacturing) principles to optimize the design methodology based on 3D printing technology.	0.5	1	-	0.5
PC2. Evaluate part consolidation, geometric freedom, and biomimicry to enhance design capabilities.	0.5	1	-	0.5
PC3. Consider key process-specific, material-driven design guidelines, rules, and considerations.	0.5	1	-	0.5
PC4. Consider limitations of 3D printing parts, pre-defined design allowables in line with industry standards to comprehend design feasibility	0.5	1	-	0.5
PC5. Include suggested material types, colors, infill levels, orientations, and finishing methods in the 3D design.	0.5	1	-	0.5
PC6. Evaluate the constraints in physical properties of materials, including supports for overhangs, walls, and corners	0.5	1	-	0.5
PC7. Determine shapes of components with fewer stress points, smaller footprints, and superior mechanical properties	1	1.5	-	0.5
PC8. Decide on the minimum level of detail required based on the specific 3D printing technology.	1	1.5	-	0.5
PC9. Quantify and prioritize necessary mechanical properties, standards, and loads for the parts.	1	1.5	-	0.5
PC10. Refer to industry standards for applicable mechanical properties, including tensile, compression, flexural strength, and wear resistance	1	1.5	-	0.5
PC11. Analyze loading conditions to design parts such that largest forces traverse planes parallel to the print bed for optimal strength	1	1.5	-	0.5

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Identify critical dimensions and features, including minimum wall thickness and support variables, adjusting for printer precision	1	1.5	-	0.5
PC13. Maximize print bed contact to minimize the need for supports and improve bed adhesion.	1	1.5	-	0.5
PC14. Orient the part optimally so the largest face lies on the print bed unless dictated otherwise by strength or geometry	1	1.5	-	0.5
PC15. Reduce supports and improve overhangs by optimizing angles to decrease printing and processing time	1	1.5	-	0.5
PC16. Avoid large flat surfaces and use rounded corners to prevent warping during printing.	1	1.5	-	0.5
PC17. Use fillets or chamfer edges to ensure smooth transitions and reduce stress concentrationS	1	1.5	-	0.5
PC18. Thicken vulnerable areas and appendages from the core to prevent breakage	1	1.5	-	0.5
PC19. Conduct operations between software systems to enhance workflows and design capabilities	1	1.5	-	0.5
PC20. Compare CAD modeling software solutions to identify the most suitable package for 3D printing technology	1	1.5	-	0.5
PC21. Incorporate AI-driven generative design methodologies and software tools for innovative solutions	1	1.5	-	0.5
PC22. Set part parameters like mount points and environmental stresses without strictly dictating the shape of the object	1	1.5	-	0.5
PC23. Extract and modify geometries generated by AI-based algorithms in CAD systems	1	1.5	-	0.5
PC24. Design and create special effects, including particle effects and dynamic simulations	1	1.5	-	0.5

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC25. Utilize software solutions to automate design workflows through scripting	1	1.5	-	0.5
PC26. Define target user needs and product requirements to ensure alignment with market expectations	1	1.5	-	0.5
PC27. Analyze product lifecycle and sustainability, considering recyclability and environmental impact.	1	2	-	1
PC28. Use feedback from prototype testing and user trials to iteratively improve product design.	1	2	-	1
PC29. Establish quality assurance protocols for testing prototypes to ensure functionality and compliance.	1	2	-	1
PC30. Ensure alignment with branding and aesthetic considerations to create products that resonate with target audiences	1	2	-	1
PC31. Establish quality assurance protocols for testing prototypes to ensure functionality and compliance with specifications	1	2	-	1
PC32. Ensure alignment with industry standards for applicable mechanical properties.	1	2	-	1
PC33. Plan for manufacturability (DfM) by considering cost-effectiveness during production	1	2	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8912
NOS Name	Design and develop finalized 3D models and product prototypes using CAD/CAM programs
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	3D Printing
NSQF Level	5
Credits	4
Version	3.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

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SSC/N8913: Structure and conduct appropriate simulation tests, design optimization techniques to validate and finalize the 3D CAD models.

Description

This unit is about conducting simulation tests and validation of the design models developed

Scope

The scope covers the following :

- Simulation Planning and Setup, Finite Element Analysis (FEA) Execution, Data Output and Analysis, Results Verification and Redesign, Topology Optimization (TO), Implementation and Continuous Improvement

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Perform multiple simulation runs to ensure the design is compatible with the chosen manufacturing processes
- PC2.** Aim to enhance the maximum pre-print accuracy of the CAD models by creating various test scenarios in iterations
- PC3.** Start with simple models and gradually modify them as the simulations present plausible results.
- PC4.** Create a detailed project plan to accommodate multiple iterations of FEA and TO (Topology Optimization) simulations coherently
- PC5.** Establish a clear set of success criteria for simulations based on project goals and performance targets to ensure results align with intended outcomes.
- PC6.** Conduct Finite Element Analysis (FEA) to understand and quantify the effects of real-world conditions on a part or assembly
- PC7.** Research and select from the most advanced FEA simulation software packages best suited for the analysis requirements.
- PC8.** Determine and prioritize various types of FEA simulations like structural, thermal, computational fluid dynamics (CFD), etc., based on the process, materials, and necessary performance indicators.
- PC9.** Generate a mesh model and refine the mesh using an iterative process of the FEA theory.
- PC10.** Define element type, choose material data set, and assign all the properties required
- PC11.** Enter sections and assign all the loads, constants, and boundary conditions according to the type of simulation
- PC12.** Request the data output according to the variables you want to study, i.e., forces, displacements, stresses, strengths, etc
- PC13.** Analyze the data by plotting contours of the requested variables, evaluating the deformed and undeformed shapes plots in the form of custom tabular reports.
- PC14.** Obtain a detailed visualization of where structures bend or twist, and the distribution of stresses and displacements as results.

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- PC15.** Utilize sensitivity analysis to understand how variations in input parameters affect simulation outcomes.
- PC16.** Verify if the upper limits of failure meet the required mechanical properties, standards, and loads as anticipated
- PC17.** Consider redesigning the models based on the results for further analysis.
- PC18.** Incorporate advanced simulation techniques such as multi-physics simulations that combine thermal, structural, and fluid dynamics analyses when necessary
- PC19.** Validate simulation results with physical testing of prototypes to verify accuracy and refine models based on observed discrepancies
- PC20.** Perform Topology Optimization (TO) after the initial FEA simulation results check out for optimal material distribution layout.
- PC21.** Select from the available list of specific TO software solutions or the usual CAD packages.
- PC22.** Conduct topology optimization using the Finite Element Method (FEM).
- PC23.** Aim to optimize material layout within a predefined set of loads, conditions, and constraints.
- PC24.** Maximize the performance and efficiency of design by removing redundant material from areas that do not need to carry significant loads.
- PC25.** Solve design challenges like reducing unnecessary weight, resonance, and thermal stresses (e.g., lattice structures)
- PC26.** Determine the minimum allowable design space necessary for shape optimization of the product
- PC27.** Use TO software to test the structural integrity and identify unnecessary material under various simulated stress scenarios.
- PC28.** Implement the design modifications continuously as indicated by the simulation test outcomes
- PC29.** Analyze cost implications associated with different design optimizations to ensure that performance improvements do not significantly increase production costs.
- PC30.** Train and mentor team members on best practices in simulation methods and software usage to enhance overall team capabilities

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Organizational policies, procedures and guidelines which relate to evaluating and developing new use cases
- KU2.** Different data sources and how to access documents and information from data sources
- KU3.** Organizational policies and procedures for documenting new developed business cases
- KU4.** Who to involve while evaluating the feasibility of new business use cases and development plans
- KU5.** The range of standard templates and tools available and how to use them
- KU6.** Knowledge of 3D simulation techniques
- KU7.** Knowledge of finite element analysis
- KU8.** Knowledge of Topology optimization software

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Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** analyze business impact and disseminate relevant information to others
- GS2.** apply problem-solving approaches in different situations
- GS3.** seek clarification and advice on problems from appropriate people
- GS4.** provide opinions on work in a detailed and constructive way
- GS5.** work effectively in a customer facing environment

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Perform multiple simulation runs to ensure the design is compatible with the chosen manufacturing processes	1	1	-	0.5
PC2. Aim to enhance the maximum pre-print accuracy of the CAD models by creating various test scenarios in iterations	1	1	-	0.5
PC3. Start with simple models and gradually modify them as the simulations present plausible results.	1	1	-	0.5
PC4. Create a detailed project plan to accommodate multiple iterations of FEA and TO (Topology Optimization) simulations coherently	1	1	-	0.5
PC5. Establish a clear set of success criteria for simulations based on project goals and performance targets to ensure results align with intended outcomes.	1	1	-	0.5
PC6. Conduct Finite Element Analysis (FEA) to understand and quantify the effects of real-world conditions on a part or assembly	1	1	-	0.5
PC7. Research and select from the most advanced FEA simulation software packages best suited for the analysis requirements.	1	1	-	0.5
PC8. Determine and prioritize various types of FEA simulations like structural, thermal, computational fluid dynamics (CFD), etc., based on the process, materials, and necessary performance indicators.	1	1	-	0.5
PC9. Generate a mesh model and refine the mesh using an iterative process of the FEA theory.	1	1	-	0.5
PC10. Define element type, choose material data set, and assign all the properties required	1	1	-	0.5
PC11. Enter sections and assign all the loads, constants, and boundary conditions according to the type of simulation	1	2	-	0.5

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Request the data output according to the variables you want to study, i.e., forces, displacements, stresses, strengths, etc	1	2	-	0.5
PC13. Analyze the data by plotting contours of the requested variables, evaluating the deformed and undeformed shapes plots in the form of custom tabular reports.	1	2	-	0.5
PC14. Obtain a detailed visualization of where structures bend or twist, and the distribution of stresses and displacements as results.	1	2	-	0.5
PC15. Utilize sensitivity analysis to understand how variations in input parameters affect simulation outcomes.	1	2	-	0.5
PC16. Verify if the upper limits of failure meet the required mechanical properties, standards, and loads as anticipated	1	2	-	0.5
PC17. Consider redesigning the models based on the results for further analysis.	1	2	-	0.5
PC18. Incorporate advanced simulation techniques such as multi-physics simulations that combine thermal, structural, and fluid dynamics analyses when necessary	1	2	-	0.5
PC19. Validate simulation results with physical testing of prototypes to verify accuracy and refine models based on observed discrepancies	1	2	-	0.5
PC20. Perform Topology Optimization (TO) after the initial FEA simulation results check out for optimal material distribution layout.	1	2	-	0.5
PC21. Select from the available list of specific TO software solutions or the usual CAD packages.	1	2	-	1
PC22. Conduct topology optimization using the Finite Element Method (FEM).	1	2	-	1
PC23. Aim to optimize material layout within a predefined set of loads, conditions, and constraints.	1	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC24. Maximize the performance and efficiency of design by removing redundant material from areas that do not need to carry significant loads.	1	2	-	1
PC25. Solve design challenges like reducing unnecessary weight, resonance, and thermal stresses (e.g., lattice structures)	1	2	-	1
PC26. Determine the minimum allowable design space necessary for shape optimization of the product	1	2	-	1
PC27. Use TO software to test the structural integrity and identify unnecessary material under various simulated stress scenarios.	1	2	-	1
PC28. Implement the design modifications continuously as indicated by the simulation test outcomes	1	2	-	1
PC29. Analyze cost implications associated with different design optimizations to ensure that performance improvements do not significantly increase production costs.	1	2	-	1
PC30. Train and mentor team members on best practices in simulation methods and software usage to enhance overall team capabilities	1	2	-	1
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8913
NOS Name	Structure and conduct appropriate simulation tests, design optimization techniques to validate and finalize the 3D CAD models.
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	3D Printing
NSQF Level	5
Credits	5
Version	3.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

SSC/N8914: Export print-ready digital design files and generate tools paths for 3D printers in the form of G-code instructions

Description

This unit is about finalising and exporting the design model to be printed in the 3D printer

Scope

The scope covers the following :

- File Preparation and Export, Slicing and Tool Path Generation, G-code Management and Optimization

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Transform the raw design files and renders into 3D printable digital file formats as Polygon Mesh models.
- PC2.** Select from different file formats to export like STL, STEP, IGES or IGS, OBJ, 3MF, or DXF, with STL being the most commonly preferred format
- PC3.** Employ appropriate methods to export based on the CAD program being used.
- PC4.** Consider the parameters used for outputting an STL file and their effect on the faceting of the mesh model.
- PC5.** Understand that smoother/less faceted surfaces (higher poly count or triangulation) result in larger STL file sizes
- PC6.** Determine factors like chordal tolerance or deviation, angle control, and minimum triangle side length.
- PC7.** Strike a balance between the model, desired surface fidelity, and the 3D printing process of choice, as 3D printing can only accept a certain size
- PC8.** Adjust the values of the chordal and angular tolerance to generate the optimal file size and mesh faceting without compromising on resolution.
- PC9.** After generating the mesh, ensure supports are created for the mesh model intelligently
- PC10.** Use mesh repair tools to check, repair, and optimize the mesh before slicing
- PC11.** Process the STL file using a slicer or slicing software to generate automated tool paths.
- PC12.** Convert the model into a series of thin layers through the slicer, producing a G-code file (.gcode) containing instructions tailored to a specific type of 3D printer
- PC13.** Determine optimal printing build parameters in the slicing software with regard to necessary mechanical properties, standards, and load requirements.
- PC14.** Input parameters like orientation, fill pattern, layer thickness, print speed, temperature, and other printer-specific settings.
- PC15.** If needed, modify model specifications partially through scaling features in the slicer before processing
- PC16.** Select the right slicing software based on organizational preferences, as the choice of slicer will impact the final print quality.

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- PC17.** Utilize advanced slicing features such as adaptive layer height for optimized print quality and reduced print time
- PC18.** Implement support structures intelligently, considering their impact on ease of removal and print quality
- PC19.** Conduct print simulations within the slicer to visualize potential issues and optimally adjust settings
- PC20.** Export G-code from slicing software and import the G-code file into the 3D printing software
- PC21.** Familiarize yourself with printer-specific G-code commands and syntax for manual adjustments if necessary
- PC22.** Analyze material flow and deposition in the slicer to ensure effective printing of complex geometries
- PC23.** Maintain a library of optimized profiles for different materials and settings for consistency in print quality
- PC24.** Conduct test prints with different settings to identify the best configurations for particular geometries or materials

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Organizational policies, procedures and guidelines which relate to evaluating and developing new use cases
- KU2.** Different data sources and how to access documents and information from data sources
- KU3.** Organizational policies and procedures for documenting new developed business cases
- KU4.** Who to involve while evaluating the feasibility of new business use cases and development plans
- KU5.** The range of standard templates and tools available and how to use them
- KU6.** Knowledge of Different 3D printers
- KU7.** Knowledge of Slicing software

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** analyze business impact and disseminate relevant information to others
- GS2.** apply problem-solving approaches in different situations
- GS3.** seek clarification and advice on problems from appropriate people
- GS4.** provide opinions on work in a detailed and constructive way
- GS5.** work effectively in a customer facing environment

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Transform the raw design files and renders into 3D printable digital file formats as Polygon Mesh models.	1	2	-	0.5
PC2. Select from different file formats to export like STL, STEP, IGES or IGS, OBJ, 3MF, or DXF, with STL being the most commonly preferred format	1	2	-	0.5
PC3. Employ appropriate methods to export based on the CAD program being used.	1	2	-	0.5
PC4. Consider the parameters used for outputting an STL file and their effect on the faceting of the mesh model.	1	2	-	0.5
PC5. Understand that smoother/less faceted surfaces (higher poly count or triangulation) result in larger STL file sizes	1	2	-	0.5
PC6. Determine factors like chordal tolerance or deviation, angle control, and minimum triangle side length.	1	2	-	0.5
PC7. Strike a balance between the model, desired surface fidelity, and the 3D printing process of choice, as 3D printing can only accept a certain size	1	2	-	0.5
PC8. Adjust the values of the chordal and angular tolerance to generate the optimal file size and mesh faceting without compromising on resolution.	1	2	-	0.5
PC9. After generating the mesh, ensure supports are created for the mesh model intelligently	1	2	-	1
PC10. Use mesh repair tools to check, repair, and optimize the mesh before slicing	1	2	-	1
PC11. Process the STL file using a slicer or slicing software to generate automated tool paths.	1	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Convert the model into a series of thin layers through the slicer, producing a G-code file (.gcode) containing instructions tailored to a specific type of 3D printer	1	2	-	1
PC13. Determine optimal printing build parameters in the slicing software with regard to necessary mechanical properties, standards, and load requirements.	1	2	-	1
PC14. Input parameters like orientation, fill pattern, layer thickness, print speed, temperature, and other printer-specific settings.	1	2	-	1
PC15. If needed, modify model specifications partially through scaling features in the slicer before processing	1	2	-	1
PC16. Select the right slicing software based on organizational preferences, as the choice of slicer will impact the final print quality.	1	2	-	1
PC17. Utilize advanced slicing features such as adaptive layer height for optimized print quality and reduced print time	1	2	-	1
PC18. Implement support structures intelligently, considering their impact on ease of removal and print quality	1	2	-	1
PC19. Conduct print simulations within the slicer to visualize potential issues and optimally adjust settings	2	2	-	1
PC20. Export G-code from slicing software and import the G-code file into the 3D printing software	2	2	-	1
PC21. Familiarize yourself with printer-specific G-code commands and syntax for manual adjustments if necessary	2	2	-	1
PC22. Analyze material flow and deposition in the slicer to ensure effective printing of complex geometries	2	2	-	1
PC23. Maintain a library of optimized profiles for different materials and settings for consistency in print quality	2	3	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC24. Conduct test prints with different settings to identify the best configurations for particular geometries or materials	2	3	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8914
NOS Name	Export print-ready digital design files and generate tools paths for 3D printers in the form of G-code instructions
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	3D Printing
NSQF Level	5
Credits	5
Version	3.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

The assessment will consist of a blend of hands-on practical evaluations, viva-voce, and online proctored scenario-based multiple-choice questions ensuring a thorough evaluation of the individual's proficiency in learning outcomes, practical understanding, and real-world application of concepts.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N8912.Design and develop finalized 3D models and product prototypes using CAD/CAM programs	30	50	-	20	100	30
SSC/N8913.Structure and conduct appropriate simulation tests, design optimization techniques to validate and finalize the 3D CAD models.	30	50	-	20	100	30
SSC/N8914.Export print-ready digital design files and generate tools paths for 3D printers in the form of G-code instructions	30	50	-	20	100	30
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	110	180	-	60	350	100

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Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.